Executive Briefing & Operations Security Summary

AUDIT SCOPE: Camera IPC56 (Rooftop Deck) | **WINDOW:** Sep 1 – Sep 30, 2026 (713.66 Continuous Hours)
DATASET: 36,198 High-Quality Face Vectors | 1,652 Discovered Identities | 179 Continuous Incident Clips

1. Executive Mission Overview

An autonomous forensic and biometric surveillance audit was conducted over 30 continuous calendar days of native 3K CCTV footage covering the RO Water Treatment Plant rooftop. The audit established continuous tracking of all personnel visiting the mechanical wing, identified shift distribution patterns, resolved critical blindspots where incident clips were prematurely truncated, and isolated every direct physical contact event against mission-critical equipment assets.

2. Key Operational Metrics

Metric Category	Audit Result	Operational Context
Total Facility Coverage	**713.66 Recorded Hours**	30 continuous calendar days (100% of Sept 2026 footage)
Discovered Personas	**1,652 Unique Identities**	Unsupervised DBSCAN cosine clustering (eps <= 0.38)
Core Operating Personnel	**12 Staff Members**	Tier 1 high-frequency staff (50+ detections each)
Direct Equipment Contacts	**18 Individuals**	Physically touched multiport valves, switches, or skid controls
Pristine Incident Clips	**179 Events (8.20 GB)**	Lossless H.265 stream cuts with +/-60s pre/post buffers
Processed Face Vectors	**36,226 Embeddings**	512-D ArcFace vectors passing strict quality threshold (Q >= 40)

3. Operational Shift Workload Distribution

Shift A (Morning, 06:00 – 13:59): 24,625 detections (68.0%) — Primary operational window for plant startup, media filter backwashing, and chemical dosing.
Shift B (Afternoon, 14:00 – 21:59): 11,188 detections (30.9%) — Secondary operating window, chemical replenishment, and shift handover checks.
Shift C (Night Vigilance, 22:00 – 05:59): 413 detections (1.1%) — Security rounds, perimeter walk-throughs, and main electrical breaker switch contacts.

4. Technical Post-Mortem & Incident Resolutions

1. **Day 1 Backwash Valve Incident (09:53:38):** Previous test cuts ended at 09:53:18, cutting off 20 seconds before operator turned the valve. *Root cause:* automated motion extraction on Clip 14 timed out after 42 seconds. *Fix:* Re-extracted full continuous 4-minute clip (Day01_0953_Manual_Incident_Audit.mp4) and formally enrolled operator as **SUBJ-1652**.
2. **Day 1 Night Guard Breaker Contact (22:27:36):** Previous cut ended at 22:27:26, missing switch contact. *Root cause:* file split at 22:28:00 prevented single-file extraction. *Fix:* Re-extracted seamless 6-minute clip (Day01_2227_Manual_Incident_Audit.mp4) spanning the boundary.
3. **Sorting Column Inversion Fix:** Rebuilt all 30 clipping indices with strict wall-clock sorting (ORDER BY timestamp ASC), eliminating timestamp inversions across all 179 incident events.

5. Management & Security Recommendations

1. Access Control: Enforce physical lockout/tagout (LOTO) padlock boxes over Zone 1A/B Multiport Valves and Zone 2 Power Switch.
2. Real-Time Spatial Alerting: Implement automated audible alerts when any unauthorized centroid breaches the 40px buffer of Zone 1 or Zone 2.
3. Forensic Storage: Retain the 179 extracted buffered MP4 files in dedicated day folders as certified evidence ground truth.